Fiscal Year:	FY 2009	**Task Last Updated:**	FY 11/26/2008
PI Name:	Freed, Lisa E M.D., Ph.D.		
Project Title:	Microgravity Tissue Engineering		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	02139-4037	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE until 11/30/2010 per PI/JSC approval notice (09/2010) NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 (jvp 6/2009) NOTE: Received NCE to 11/30/2009 (from 11/30/2008) per PI (11/08) NOTE: Received NCE to 11/30/2008 (from 11/30/2007) per P. Denkins/JSC (7/07) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vunjak-Novakovic, Gordana (Columbia University) Rupnick, Maria (Harvard Medical School) Guilak, Farshid (Duke University Medical Center) Padera, Robert (Harvard Medical School)		
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Task Description:

Our research involves cell- and tissue-levels studies of 3D engineered tissue constructs and utilizes models of relevance to tissue loss in astronauts in microgravity and skeletal and myocardial tissue regeneration in human patients on Earth. We selected to study skeletal and cardiac tissue regeneration because: (a) spaceflight adversely affects these tissues, (b) improved treatments are needed for millions of patients who suffer from progressive skeletal and cardiac diseases, (c) these two distinctly different tissue types can provide paradigms for studies of individual and interactive effects of biophysical stimuli, and (d) we and others have already demonstrated the feasibility of engineering skeletal and cardiac-like tissue, and thus can capitalize on recent advances to promote groundbreaking research and technology transfer. We previously designed in vitro models based on cells, 3D biomaterial scaffolds, and advanced bioreactor systems, and showed bioreactor-grown constructs were larger and functionally superior to controls grown in petri dishes. The current limitations of our tissue engineering research are as follows: (i) long periods of in vitro culture are required to generate mechanically functional engineered skeletal tissue, (ii) cardiomyocyte viability, differentiation, and contractility are poor in engineered cardiac tissue, and (iii) spaceflight tissue culture studies are limited by the lack of proper controls necessary to derive underlying mechanisms. We are addressing these limitations by: (a) integrated studies in cell and molecular biology, mass and momentum transport, design and operation of bioreactors including microgravity analog systems, tissue physiology and biomechanics, and implant biocompatibility and remodeling, (b) experiments interpreted with the aid of mathematical models, (c) complementary in vitro and in vivo studies, and (d) functional evaluation at various hierarchical scales, i.e. molecular, cell, and tissue.
The project is organized as two Aims: (1) skeletal tissue engineering, and (2) cardiac tissue engineering. The global hypothesis is that biophysical stimuli that are physiologic in nature, magnitude and regime of application can direct cells to regenerate functional tissues in vitro. We can test this hypothesis in controlled studies of cells cultured on 3D scaffolds in bioreactors and in the presence of biophysical stimuli. The work is expected to help define how biophysical cues regulate skeletal and cardiac tissue development and function, and yield engineered skeletal and cardiac tissue constructs with superior functionality compared to previous reports.

Our studies of skeletal and cardiac tissue constructs involve the cultivation of cells on 3D scaffolds in advanced bioreactors and in the presence of biophysical stimuli. Critical research areas include cell sourcing (cell expansion, stem cell differentiation), development of novel biomaterial scaffolds and advanced bioreactors for cell seeding of scaffolds and construct cultivation in the presence of biophysical stimuli (e.g. controllable hydrodynamic shear, cyclic stretch, growth factors, pharmaceutical agents), and construct implantation in vivo for assessment of cell survival and graft integration, and capacity for defect repair. Tissue engineered constructs are assessed with respect to structure, molecular composition, and mechanical and electrophysiological function in order to evaluate individual and interactive effects of experimental variables. Key equipment includes facilities for working with cells, tissue specimens, experimental animals, polymer processing, bioreactor systems, biomechanical and electrophysiological test systems, and analytical instruments for histology, biochemistry, molecular biology, microscopy and image analysis. Construct properties are correlated with in vitro culture conditions and the outcome of in vivo studies, in order to provide feedback necessary to tailor model system parameters and obtain the best overall result. The expected outcomes are improved understanding of how biophysical stimuli regulate skeletal and cardiac tissue development and function, and the generation of tissue engineered constructs with functionality superior to that previously reported in the literature.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our research is relevant to the Bioastronautics Roadmap questions #2) impaired fracture healing, #3) injury to joints, #5) occurrence of serious cardiac dysrhythmias, #6) diminished cardiac function, #19) pharmacology of space medicine, and possibly also to #21) rehabilitation on Mars. Our combined use of 3D engineered tissue constructs and advanced bioreactor systems provides a unique model system for controlled in vitro studies at the cell and tissue-levels that may be relevant to treating skeletal and myocardial tissue loss and dysfunctions in astronauts. Biophysical stimuli play a critical role in tissue growth, maintenance, and remodeling. During human spaceflight, altered biophysical stimuli and the resulting systemic adaptations can cause fluid and electrolyte imbalance, altered neurohormonal regulation, diminished tissue remodeling, and loss of bone and muscle mass. Exercise regimens and pharmaceutical interventions have been proposed to try to maintain skeletal and cardiac tissues in-flight, and to try to rehabilitate deconditioned skeletal and cardiac tissues post-flight. However, the ability to determine the underlying mechanism(s) or assess the potential risks/benefits of currently used countermeasures is limited by a lack of experimental models. Our controlled in vitro studies of how 3D tissue engineered constructs respond to biophysical stimuli are expected to help identify how the microgravity environment of space may compromise the ability of skeletal and cardiac tissues to respond to and/or withstand biophysical stimuli. Improved understanding can reasonably be expected to aid in the development of better countermeasures.
Our tissue engineering research has implications not only for astronauts but also for humans on earth, e.g. by developing in vitro analogs for experimental pharmaceutical studies, and new treatments for skeletal and cardiovascular diseases. Skeletal and cardiac tissues perform structural and mechanical functions that are vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over 40 million Americans suffer from arthritis, which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited capacity for self-repair, and no currently available therapy can predictably restore a functional articular surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle, and heart disease is the leading cause of morbidity and mortality in developed countries. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting 3D tissue constructs that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, despite many advances, tissue engineered grafts for cartilage and cardiac regeneration still face many challenges in order to be of widespread clinical use, hence, this tissue engineering research can clearly benefit life on Earth.

To date, this NASA research project has been highly productive, yielding more than 90 publications in leading peer-reviewed scientific journals including the one-and-only long duration microgravity tissue engineering experiment carried out aboard STS-79, Mir Space Station and STS-81, and published by Freed et al., Proceedings of the National Academy of Sciences USA. Also, this NASA research project resulted in over 100 abstracts and six patent applications, and has helped to train seventeen postdoctors, nine graduate students including six who earned PhD degrees, and numerous undergraduates. Professional activities of the co-investigators include teaching of undergraduate and graduate students chairing and organizing scientific conferences, and serving on scientific research panels and scientific and editorial advisory boards. NASA used the data and experience gained in our ground and flight studies to develop two bioreactor systems: (i) Advanced Rotating Bioreactor System and (ii) Cell Culture Unit. The Co-investigators use research supported by this NASA project as examples in the courses that we teach. For example, L.E. Freed has taught graduate-level courses in the Harvard-MIT Division of Health Sciences and Technology, e.g. "Biomaterials and Tissue Engineering for Medical Devices" (HST-521), and F. Guilak has taught courses at Duke University and the Center for Biomolecular and Tissue Engineering., e.g. "Introduction to Biologically Inspired Materials" (MEMS-265/BME-265). Hence, this research has had a major impact on America with Earth benefits.

As itemized in Section 6 of this report and summarized briefly below, our Task Progress during this reporting period includes 8 published research articles, 2 in-press research articles, and 1 patent application. Our recent Nature Materials E-publication was spotlighted on the MIT homepage, and featured in several articles in the scientific and popular press. The patent application is the subject of an NASA Tech Brief article that is currently in-press.
Bioreactor development.

Perfusion bioreactors are known to enhance the regeneration of functional 3D engineered tissues. In this context, we designed, fabricated, and tested a new oscillatory perfused bioreactor that can provide effective convective-diffusive mass transport of gases, nutrients, and regulatory molecules to developing tissue constructs in the context of a commercially appealing device [Moretti, Freed, Langer, Patent Application, 2008].

Skeletal Tissue Engineering.

Skeletal tissue engineering is currently limited by failure of traditional 3D scaffolds to mimic the complex mechanical and structural

Task Progress:	properties of native skeletal tissues. Using a 3D woven poly(?-caprolactone) (PCL) scaffold with cartilage-mimetic mechanical properties, we demonstrated spatially uniform deposition of cartilaginous extracellular matrix by cultured human mesenchymal stem cells (hMSC) within a relatively short (3 week) period of in vitro culture [Valonen et al., ORS Abstract]. Different culture media supplements and tissue culture vessels were compared. Mechanical testing showed that the PCL scaffold was responsible for the measured construct mechanical properties. These findings demonstrated that cartilage grafts based on a 3D woven PCL scaffold have the potential to provide mechanical function shortly after cell seeding, instead of requiring prolonged cultivation in vitro. In another study, we used a 3D porous silk scaffold and the NASA rotating bioreactor (STLV) to study osteogenesis by cultured hMSC over 3-6 weeks of culture. [Augst et al., J. Royal Soc. Interface, 2008]. Microcomputerized tomography (μ-CT), gadolinium-enhanced magnetic resonance imaging (dGEMRIC), and histomolecular assays of the resulting constructs demonstrated that bone formation progressed to a markedly larger extent than cartilage formation in this model system. In another study, we validated a bioreactor designed to provide shear stress and/or hydrostatic pressure to engineered cartilage constructs [Moretti, Bio-Medical Materials and Engineering, in press]. Using a non-woven scaffold and bovine calf chondrocytes, we compared three groups: (i) perfusion combined with hydrostatic pressure (10 MPa, 1 Hz, 4h/day), (ii) perfusion without hydrostatic pressure, and (iii) static control cultures. Cell viability was highest in constructs cultured with both perfusion and hydrostatic pressure, and cartilaginous matrix deposition was higher constructs cultured with perfusion compared to static controls. These findings demonstrate that application of mechanical stimuli to developing engineered cartilage constructs can elicit specific responses and improve the quality of the resulting tissues. Cardiac Tissue Engineering Cardiac tissue engineering is currently limited by: (i) inadequate survival, alignment, and differentiation of component cardiomyocytes, and (ii) failure of traditional 3D scaffolds to mimic the complex mechanical and structural properties of native cardiac tissue. Using an oscillatory perfused bioreactor we showed that slow, bi-directional perfusion and insulin-like growth factor-I (IGF) independently and interactively enhanced the survival, differentiation and contractile performance of engineered cardiac constructs [Cheng et al., Tissue Engineering 2008, Aug. 31 E-Pub]. Perfusion enhanced cell viability, contractile amplitude and excitation threshold, and also increased construct amounts of cardiac troponin-I, connexin-43, and total protein. Interestingly, myofibrillogenesis and spontaneous contractility were present only in constructs cultured with perfusion, although contractility was inducible by electrical field stimulation of grafts from all groups. These findings demonstrate that multi-factorial stimulation of engineered cardiac tissue by perfusion and IGF independently and interactively affected heart cell survival, differentiation, and contractility. In another study, we used a 3D porous poly(glycerol-sebacate) (PGS) scaffold with elastomeric properties to show that sequential co-culture of cardiac fibroblasts and cardiomyocytes enhanced cell attachment and differentiation, and construct contractile function. Also, we demonstrated feasibility of implanting cell-free porous PGS scaffolds on the surface of infarcted myocardium in a rodent model [Radisic et al., J. Biomed. Mater. Res., 2008]. In a related study, we used 3D co-cultures of cardiac fibroblasts and cardiomyocytes and media supplemented or not with the matrix metalloproteinase (MMP) inhibitor doxycycline to show that the remodeling potential of engineered cardiac constructs is MMP-mediated [Nichol et al., Biochem. Biophys. Res. Comm., 2008]. In an explicit attempt to mimic the structural and mechanical properties of native myocardium, we used excimer laser microablation to create ‘accordion-like’ honeycomb scaffolds with directionally dependent structural and mechanical properties [Engelmayr et al., Nature Materials, 2008 Nov. 2 E-Pub]. This scaffold had three principal advantages as follows. First, its mechanical properties closely matched those of native heart tissue, e.g. it was stiffer when stretched circumferentially as compared to longitudinally. Second, constructs created by culturing neonatal rat heart cells on the scaffold exhibited directionally dependent electrophysiological properties similar to native tissue. In other words, when an electrical field was applied the construct contracted more readily in one direction than in another. In a third advantage, the scaffold exhibited an intrinsic ability to guide the orientation of cultured heart cells. As the microfabricated scaffolds used in these experiments were too thin to address reconstruction of full-thickness myocardium, we also created new honeycomb scaffolds that, among other things, allowed much thicker, multi-layered tissue structures.

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